

RAAHE LAPALUOTO, Finland

WMO: 028720

Lat: 64.6658N

Lon: 24.4069E

Elev: 7

StdP: 14.69

Time Zone: 2.00 (EUE)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-9.7	-4.0	-14.8	2.5	-9.6	-8.7	3.5	-3.9	41.8	31.4	39.3	31.0	7.2	70	0.808

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.2	75.9	65.3	72.2	63.9	69.1	62.1	67.6	73.6	65.2	70.6	63.1	67.9	10.0	10

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.4	94.0	70.6	63.0	86.0	68.4	60.8	79.7	66.3	32.0	73.5	30.2	71.0	28.6	68.0	75.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
35.0	30.8	27.2		-13.0	82.0	5.9	5.8	-17.2	86.2	-20.7	89.6	-24.0	92.9	-28.2	97.1	
			DB	-12.1	69.6	7.2	4.0	-17.3	72.4	-21.5	74.8	-25.5	77.0	-30.8	79.9	
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	39.4	18.3	20.3	25.4	34.2	46.3	54.6	61.7	58.9	52.0	40.3	33.0	26.5
	DBStd	16.61	11.31	12.05	8.63	5.53	7.77	6.12	5.90	5.14	5.14	5.86	7.40	10.22
	HDD50	4823	984	831	763	475	167	21	1	2	38	304	510	727
	HDD65	9400	1449	1251	1228	924	582	318	138	201	390	767	960	1192
	CDD50	956	0	0	0	1	53	159	364	278	99	2	0	0
	CDD65	58	0	0	0	0	3	6	37	12	0	0	0	0
	CDH74	229	0	0	0	0	13	27	144	45	0	0	0	0
	CDH80	52	0	0	0	0	1	7	35	9	0	0	0	0
	WSAvg	13.5	14.2	14.3	14.4	12.3	12.7	12.7	11.0	12.3	13.3	14.2	14.8	15.8
	PrecAvg	21.4	1.5	1.2	1.1	1.1	1.7	2.0	2.8	2.4	2.0	2.0	1.9	1.7
(15) Precipitation	PrecMax	26.2	2.7	2.2	2.2	2.6	3.2	4.4	5.2	4.5	4.0	3.4	3.7	3.7
	PrecMin	15.9	0.4	0.3	0.2	0.4	0.5	0.2	0.6	0.7	0.7	0.2	0.3	0.5
	PrecStd	3.1	0.7	0.6	0.5	0.6	0.7	1.1	1.1	1.1	1.0	0.9	0.8	0.7
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	37.3	38.5	43.1	54.3	75.6	77.2	85.1	79.6	68.3	54.6	47.3	41.6
		MCWB	36.2	36.5	36.7	48.6	57.9	64.8	66.7	67.4	59.6	51.6	44.9	40.0
	2%	DB	35.4	36.8	38.7	48.2	67.8	70.6	77.3	72.5	63.7	52.0	45.5	39.3
		MCWB	34.0	34.9	35.5	42.6	55.7	60.4	66.9	65.2	57.7	49.3	43.9	37.4
	5%	DB	34.3	35.5	37.1	43.9	61.9	66.9	73.9	68.9	60.8	49.6	43.3	37.8
		MCWB	33.2	33.8	34.6	39.5	53.4	59.1	65.4	62.9	56.5	47.3	41.8	36.1
	10%	DB	32.8	34.0	35.5	41.5	57.5	64.0	70.8	66.3	58.8	48.1	41.5	36.3
		MCWB	31.9	32.7	33.4	38.0	50.2	57.3	64.1	61.3	55.5	45.8	40.1	34.9
	0.4%	WB	36.6	37.0	38.5	50.1	59.9	65.9	71.5	69.2	60.9	52.7	45.7	40.2
		MCDB	36.9	38.3	41.7	53.5	70.1	72.6	76.3	76.1	66.1	54.0	46.7	41.1
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	34.4	35.2	36.2	42.9	57.0	62.5	68.8	66.3	58.9	49.9	44.0	37.7
		MCDB	35.1	36.4	38.1	47.5	65.9	68.4	74.6	71.8	62.3	51.4	45.3	38.8
	5%	WB	33.3	34.1	34.9	40.3	54.0	60.1	66.8	63.8	57.3	47.7	42.1	36.4
		MCDB	34.1	35.2	36.6	43.2	61.7	65.9	72.5	68.1	59.8	49.2	43.2	37.5
	10%	WB	31.9	32.9	33.8	38.3	51.0	57.9	64.7	61.9	56.0	46.1	40.3	35.1
		MCDB	32.9	34.0	35.2	40.9	56.5	63.1	70.0	65.6	58.5	47.8	41.4	36.1
(35) Mean Daily Temperature Range	5% DB	MDBR	9.4	9.0	9.8	8.6	10.1	9.3	9.2	9.2	8.2	7.1	5.9	6.9
		MCDBR	5.6	6.3	7.3	12.5	17.4	15.3	15.1	13.3	12.2	7.7	6.2	5.6
		MCWBR	5.6	5.6	5.6	8.7	9.7	8.7	8.3	8.2	8.0	6.6	6.1	5.1
	5% WB	MCDBR	6.1	6.3	6.8	11.8	16.1	12.9	13.1	11.1	9.8	7.3	6.0	5.4
		MCWBR	6.2	5.7	5.6	8.7	9.7	8.3	8.3	7.7	7.2	6.7	6.2	5.0
(40) Clear-Sky Solar Irradiance	taub	0.213	0.279	0.308	0.346	0.347	0.351	0.370	0.355	0.332	0.310	0.343	0.352	
	taud	1.967	2.115	2.163	2.222	2.424	2.479	2.446	2.480	2.525	2.362	2.539	2.585	
	Ebn at Noon	99	192	242	260	273	275	265	258	240	186	91	41	
	Edh at Noon	8	19	29	34	31	30	30	27	21	16	7	4	
(44) All-Sky Solar Radiation	RadAvg	38	207	673	1199	1580	1767	1623	1212	674	246	53	12	
	RadStd	3	40	69	97	153	145	149	114	69	40	9	1	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (39 neighbors)	+0.86	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page